

Interaction Design of ASD Children's Online Health Community Based on Social Network Structure and User Behavior Analysis

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Abstract

Objective: To summarize the characteristics of existing online health community interactions for children with autism and use them to optimize the community interaction service model. **Method:** Based on the social network analysis theory and natural language processing theory, software such as Gephi and Python were used to analyze the user interaction system from four dimensions: overall network structure, individual network structure, word frequency topic tendency, and sentiment tendency. **Results:** Get the social network structure and text features. **Conclusion:** Proposing an interactive system design for an online health community platform for children with autism centered on opinion leaders.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing design and evaluation methods; Social network analysis.

Keywords

Online Health Community, Baidu "Autism" Bar, Interactive Systems, Social Network Analysis Theory, Natural Language Processing Theory

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1 INTRODUCTION

Autism Spectrum Disorder (ASD) is a common neurodevelopmental disorder, characterized by social communication impairments and repetitive behaviors^[1]. In China, the prevalence of ASD is 1.0% to 1.2% among newborns^[2]. Over 10 million people have autism, including more than 2 million children aged 0-14^[3]. However, the development of services is limited by a shortage of specialists, poor educational resources, lack of standards, and inadequate service

coverage^[4]. Additionally, the uneven distribution of medical resources hinders access to long-term treatment, especially in rural areas. The "Internet + Healthcare" initiative has rapidly expanded mobile healthcare services^[5]. This integration has improved rehabilitation options for children with ASD, overcoming the time and geographic barriers of traditional healthcare models^[6].

In related studies, Zhang Lie and colleagues applied service system design principles to propose online rehabilitation services^[4]. Wang Peiyao analyzed the interactions of parents in the "Autism Bar" forum, focusing on their emotional expression, social motivations, and needs^[7]. Ni Zhenni and her team mapped the knowledge-sharing and misinformation networks in the forum, proposing strategies to limit false information spread^[8]. Deng Jun and colleagues developed a model of opinion leaders within online health communities, using social network analysis to study network structures and user interactions^[9]. Current research on autism platforms overemphasizes opinion leaders, neglecting broader user interactions, interdisciplinary approaches, and practical applications. This creates a theory-practice gap in developing user-centered online health systems.

This paper investigates the interaction system of online health communities for children with autism, using Baidu's "Autism Bar" as a case study. It analyzes user interactions and behavioral patterns using social network and text mining methods. This study evaluates current platforms, proposes improvements, and applies theoretical insights to guide system design practice. The paper structure is shown in Figure 1.

2 Theoretical Foundation

2.1 Social Network Analysis Theory

Originating in 1950s sociology research [10], Social Network Analysis (SNA) models social systems as node-link networks, where nodes represent actors (e.g., individuals or organizations) and edges denote their relationships. It mathematically abstracts these connections [11] to reveal structural patterns, node-level dynamics, and interaction processes, thereby offering insights into network behavior, information diffusion, and influence propagation.

2.2 Natural Language Processing Theory

Natural Language Processing (NLP) is an interdisciplinary frontier field that combines computer science, artificial intelligence, and linguistics^[12]. This method uses computational techniques to mine information and analyze data from natural language content in text, obtaining results such as semantic features, sentiment tendencies,

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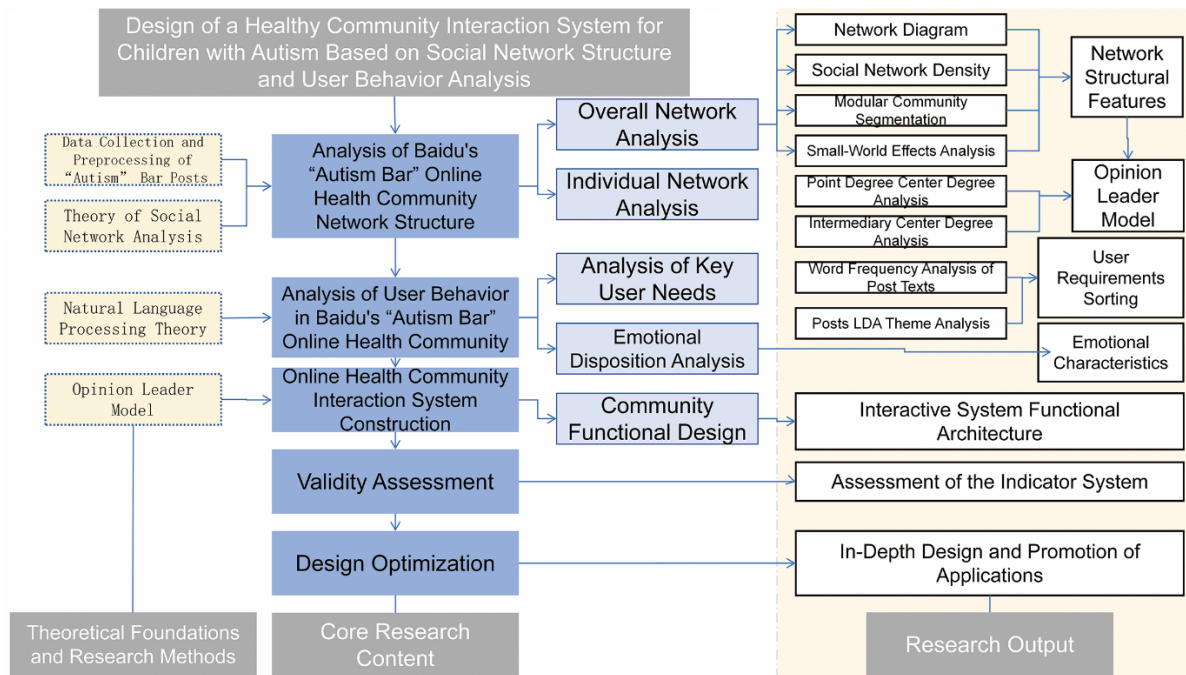


Figure 1: Thesis Framework Diagram

and category classifications. It then infers the emotions and areas of interest embedded within the text.

3 Research Subject

As a complex information ecosystem, online health communities place users at the core of the ecosystem^[6], with a series of intricate and complex interaction behaviors between users forming a complete social network. Based on the aforementioned theoretical foundation, for the social network formed by the user group in the online health community for children with autism, social network analysis is applied. Data processing is carried out using software such as Ucinet and Gephi to analyze its network structure. For the textual information involved in the online interactions of the user group, Python is used to extract and conduct text analysis.

In the existing online health community for children with autism, Baidu Tieba's "Autism Bar" was selected. Baidu Tieba is the largest Chinese-language communication community in the world, with a clear thematic focus. It gathers users interested in the same topic for online interactions. This platform is a grassroots, low-threshold channel that provides universal access to information without requiring registration or payment. It has become the preferred source of support for economically disadvantaged families. It has been created for the longest period of time, with extensive and rich user data, and can be studied as representative of a broad ASD online health community. In this study, a Python crawler program was used to collect data from the topic posts in "Autism Bar." The topic posts in the forum are ranked based on the descending order of the latest replies, meaning that the posts ranked higher have greater attention, more discussion, and stronger timeliness. The data, collected as of October 6, 2024, included user interaction data from

3,000 topic posts on the homepage of "Autism Bar," with the earliest post dating back to January 1, 2011. The collected data specifically included the poster's username, post content, posting time, replying user's username, reply content, and reply time. The users in "Autism Bar" are treated as nodes, and the reply relationships between users are regarded as edges connecting the nodes. The number of replies is used as the value to assign to each edge, and these elements are integrated into a directed multivalued interaction network. To study the overall network and the interaction behavior relationships between users, 1,588 posts with no replies were removed, and redundant, meaningless reply data were deleted. The processed data involved 13,151 users and a total of 82,660 reply entries. The preliminary processed data were subjected to social network analysis and natural language processing analysis. A matrix list was used to represent the "reply" relationships between users. Software such as Ucinet and Gephi was used for data computation and visualization, analyzing the interaction network graph data between user groups to summarize the network structure features. Additionally, Python was used for text data cleaning, frequency processing, and generating word cloud diagrams, as well as summarizing topic classification and sentiment tendencies.

4 Social Network Structure Analysis of Autism Online Health Community

In applying social network theory to the "Autism Bar" platform, each user is modeled as a node, and comment-reply interactions are transformed into directed edges linking users. Visualization and computational analysis are conducted using software tools, examining both overall and individual network levels.

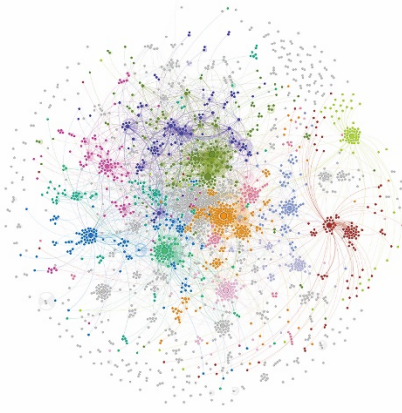


Figure 2: Baidu “Autism Bar” Network Structure Diagram

4.1 Overall Network Structure Analysis of Baidu “Autism Bar”

The overall network comprises all relational ties among nodes within a defined boundary. Using Gephi, user interactions in “Autism Bar” are visualized to generate the overall network structure (Figure 2), applying a force-directed algorithm to represent relational proximity. Node size reflects interaction frequency, increasing with the number of connections. A modularity algorithm partitions the network into smaller communities based on relational closeness; the top twelve, ranked by node count, are color-coded.

The diagram shows that node size and color are generally correlated: users with more social ties tend to appear at the centers of smaller community clusters, while larger communities are positioned nearer the center of the overall network. This layout effectively illustrates the interaction patterns within “Autism Bar.” The following sections analyze the network from three perspectives: density, community structure, and the small-world effect.

1) Network Density of Baidu “Autism Bar” Social Network.

Network density measures the closeness of node connections and is commonly used to assess network integrity. It is defined as the ratio of actual to possible edges, ranging from 0 to 1. A complete graph, in which every pair of nodes is connected, has a density of 1. Let L represent the number of actual edges, N represent the number of nodes, and the density $d(G)$ is calculated using the following formula:

$$d(G) = \frac{L}{N(N-1)} \quad (1)$$

The calculated overall network density of Baidu’s “Autism Bar” is 0.001, indicating extremely sparse user relationships and weak interaction frequency. Direct connections between nodes are limited, with a short and fragmented relationship chain, constraining information dissemination and leading to a decentralized distribution of community resources. This environment facilitates the formation of small groups and the emergence of opinion leaders due to uneven influence distribution. Although low density implies limited cohesion, it also allows for greater flexibility and the integration of diverse viewpoints. Notably, 29,302 users (35.45% of all participants) posted only once and exhibited no interaction with others, often

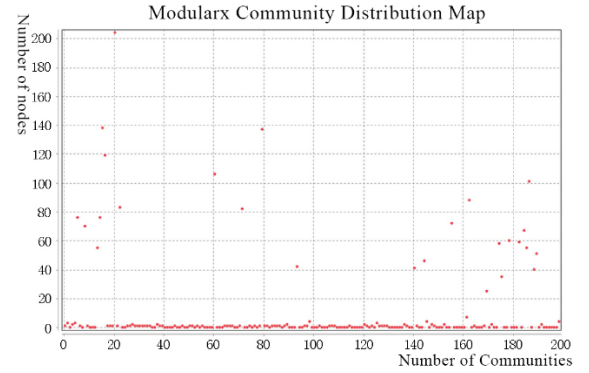


Figure 3: Modular Community Distribution Diagram

replying to themselves. These isolated users—typically on the network periphery—align with the low-engagement “tourist” profile, weakening network integrity. When these users are excluded and the density is recalculated, the value rises to 0.025, supporting this conclusion.

2) Modular Community Division of Baidu “Autism Bar”.

Modular community division segments a social network into smaller subgroups based on node similarity, each with distinct structural and functional features. Modularity values range from 0 to 1, with higher values indicating more clearly defined community structures. Let m represent the number of edges in the social network, A_{ij} represent the node pairs in the adjacency matrix A , and k_i and k_j represent the degrees of nodes i and j , respectively. The formula is as follows:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(i, j) \quad (2)$$

As shown in Figure 3, incorporating edge weight ratios into the calculation yields a modularity value (Q) of 0.721, identifying 199 communities. After excluding groups with fewer than 10 members, 26 substantial subgroups remain. A modularity of 0.721 indicates a well-defined community structure, with dense intra-community connections and sparse inter-community links. The high quality of division reflects strong similarity and cohesion among nodes within each community.

3) Small-World Effect Analysis of Baidu “Autism Bar”. The small-world effect describes networks with a short average path length and a high clustering coefficient. Using Gephi, a random network of similar size yielded a clustering coefficient of 0.02 and an average path length of 5.939. In contrast, the “Autism Bar” network showed a clustering coefficient of 0.302, indicating significantly stronger local clustering. This suggests the formation of tightly knit user groups and a pronounced community structure.

The formula for calculating the average path length is as follows:

$$L = \frac{1}{\frac{1}{2}N(N-1)} \sum_{i \neq j} d_{ij} \quad (3)$$

where d_{ij} represents the number of paths between two nodes, i and j . The average path length of the “Autism Bar” network is 3.67, indicating that any two users can connect via four intermediaries. This short path length reflects efficient information dissemination and strong network influence. Combined with the compact structure,

Table 1: Centrality Ranking

User	Out-Degree	In-Degree	Standardized Out-Degree	Standardized In-Degree	Standardized Centrality
Zhuji Rudaoshi	65	1263	0.012	0.238	0.25
Ice Cream Cat	884	201	0.166	0.038	0.204
Catch the Panda	5	882	0.001	0.166	0.167
Cige	8	845	0.002	0.159	0.161
Use_5XW25ZU	815	7	0.153	0.001	0.154
Waiting for the Sunflower	801	12	0.151	0.002	0.153
Easy Daoist	618	163	0.116	0.031	0.147
me_No Place 8 Cubs	14	683	0.003	0.129	0.132
Ardell	354	300	0.067	0.056	0.123
Use_QeSb5SQ	22	615	0.004	0.116	0.12

it suggests that nodes are relatively proximate, forming a tightly connected network.

The high clustering coefficient and short average path length indicate that the network exhibits both local clustering and global connectivity, characteristic of a small-world effect. This structure facilitates user interaction, information dissemination, and community formation.

4.2 Individual Network Structure Analysis of Baidu “Autism Bar”

An individual network is a localized network centered on a specific node, including only nodes that interact with it. Irrelevant nodes are removed, and the node’s individual attributes are analyzed. This study examines individual network properties through degree centrality and betweenness centrality.

1) Degree Centrality Analysis. Degree centrality reflects the number of nodes directly connected to a given node. In a directed network, it is divided into out-degree and in-degree. Out-degree represents the number of outgoing edges, i.e., the number of replies or comments a user has made. In-degree represents the number of incoming edges, i.e., the number of replies or comments received by the user. The standardization formula for out-degree and in-degree is as follows:

$$C'_d(v_i) = \frac{C_d(v_i)}{n-1} \quad (4)$$

Out-degree and in-degree calculations were performed for all nodes. Table 1 presents the data for the top ten users ranked by centrality. Higher values indicate greater user activity. These users influence the formation and development of network public opinion. No direct proportional relationship exists between out-degree and in-degree. Users with a high out-degree actively disseminate information, engaging in one-on-one exchanges that facilitate precise communication. They drive community interaction, often serving as maintainers, contributing to rule creation, dispute mediation, and overall community stability. In contrast, users with a high in-degree, supported by a large following, can quickly reach a broad audience, accelerating information transmission. Their influence extends to guiding community discussions, shaping its culture, and impacting long-term development and user retention.

2) Betweenness Centrality Analysis. Betweenness centrality reflects a node’s ability to act as a “bridge” connecting other nodes in the network. It is measured by the frequency of the node’s appearance on all shortest paths. The calculation involves the number of shortest paths between two nodes, s and t , that are not directly connected, and the number of shortest paths that pass through node v . The formula is as follows:

$$g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (5)$$

The users were ranked based on their calculations, and the top ten are shown in Table 2. The average value calculated is 6703.941, with approximately 17.1% of users exceeding the average. This indicates that only a small portion of users in the entire social network act as information intermediaries and have a certain informational advantage. Most users only serve the role of information inflow or outflow. This suggests that power or control within the network is relatively centralized, which helps maintain the stability and order of the network because key users are more effective at managing the network, responding to crises, or solving problems. On the other hand, this centralized power structure also makes the network more fragile, as once these key nodes or individuals are attacked, fail, or lose control, the entire network may be severely impacted or even collapse.

The social network analysis of Baidu’s “Autism Bar” revealed a network structure characterized by central density and peripheral sparsity, reflecting the small-world effect. Information flow is strong, and user interactions are frequent. However, power distribution is uneven, with discourse power concentrated among a few key users. These users hold influential positions in the network, aiding community management and information flow. However, individual instability and users’ subjective biases can distort information, undermining the network’s stability.

5 User Behavior Analysis of Baidu “Autism Bar”

Text data analysis was conducted on the posts and comments of users in the topic threads. The analysis focused on three dimensions: word frequency and sentiment values, to examine users’ social

Table 2: Betweenness Centrality Ranking

Rank	User	Betweenness Centrality	Standardized Value
1	ZhuZi Rudaoshi	439703.156	49.29
2	Dream’s Third Layer	230418.078	25.829
3	Pig	230352.078	25.822
4	Liu Wencai’s Grandfather	218061.078	24.444
5	Ardell	189443.906	21.236
6	Use_7GCaQXA	146084.609	16.376
7	Ice Cream Cat	117816.156	13.207
8	Cloud Deep Wind Shelter	117147.273	13.132
9	Ren Qi Xiawa	117081.273	13.125
10	Zhao Ge Wan Jiu	112681.063	12.631



Figure 4: Word Cloud Diagram

behaviors, identify their areas of interest, and assess their social emotions.

5.1 Word Frequency Topic Analysis of Baidu “Autism Bar”

1) Text Word Frequency Analysis of Topic Posts. The comment content was tokenized using the Jieba program for text analysis. Python was used to preprocess the text dataset by removing unnecessary characters such as emojis, advertisement markers, invalid links, and addresses. Redundant text was removed through deduplication and compression, and short texts with no meaningful content were discarded. Finally, the Jieba word segmentation library was used to tokenize the Baidu Tieba text dataset. The Harbin Institute of Technology stopword dictionary and Baidu stopword dictionary were used to remove stopwords, creating a base corpus. The corresponding TF-IDF values were calculated to observe the contribution of each term to the document’s theme. High-frequency and low-distinctiveness terms were filtered out to obtain the document’s feature vector. The top 100 high-frequency words were selected to generate a word cloud, as shown in Figure 4.

Through the calculation and analysis of the TF-IDF word frequency values, the main areas of focus for the user group within

Baidu Tieba can be identified. The most discussed topics are related to institutions, such as exchanges and comparisons of various institutions. This is due to the fact that families of children with autism often face limited access to support channels, and the autism social support system is still underdeveloped. As a result, these families have difficulty finding suitable medical institutions for professional diagnosis and systematic intervention. Therefore, in the process of seeking online assistance, the area of greatest concern for users is the introduction and evaluation of institutions.

2) Topic Post Text LDA Topic Analysis. The LDA (Latent Dirichlet Allocation) model is an analysis tool specifically designed to generate topic vocabulary by identifying the latent themes in the text and their interrelationships. This helps reveal the intrinsic thematic structure of the text data. The LDA topic model was used to analyze the comment texts in “Autism Bar,” and the “Autism Bar” was ultimately divided into four topic modules. The characteristic words for each topic are shown in Table 3.

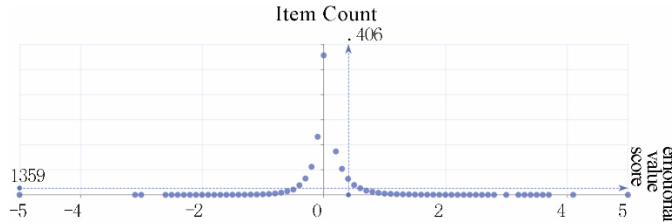
The four topics focus on the following: (1) The “Self-Recording” module, characterized by the detailed recording of personal intervention treatment experiences and emotional journeys, which includes vocabulary describing the process, progress, and feelings of personal intervention and treatment. (2) The “Emotional Support” module, aimed at expressing emotions, seeking emotional resonance, and psychological support, which involves vocabulary related to expressing emotions and seeking comfort or resonance. (3) The “Theoretical Knowledge” module, focused on sharing professional medical pathological knowledge, research findings, and insights, which includes medical terms, pathological knowledge, research results, etc. (4) The “Medical Pathways” module, which focuses on discussing intervention methods, evaluating and recommending medical institutions, and other practical information, involving medical institutions, treatment methods, expert advice, etc. This result reveals the key areas of focus and the main objectives of social interaction for users in the autism health community.

3) Sentiment Tendency Analysis of Baidu “Autism Bar”. A sentiment analysis model was trained using Python programming, and a sentiment dictionary containing both positive and negative words was created. Each word was matched with a sentiment value, with positive sentiment words assigned positive values, negative sentiment words assigned negative values, and neutral words assigned a value of zero. The values range from -5 to 5. Based on the sentiment values of the words and their weight in the text (such as word frequency, position, etc.), the scores in the sentiment dictionary were calculated for the tokenized word list, determining the overall sentiment tendency of the text. The sentiment scatter plot is shown in Figure 5.

In the comment texts of “Autism Bar,” negative emotions make up the largest portion at 42.73%, followed by positive emotions at 29.35%, and neutral emotions at 27.92%. This indicates a significant level of anxiety and concern among the parents and followers of children with autism. The prevalence of negative emotions has contributed to a negative atmosphere within the community, which may hinder the stable development of the online health platform. The sentiment scatter plot demonstrates the distribution of emotional data in the text. The horizontal axis represents the sentiment value, while the vertical axis shows the number of comments at

Table 3: Characteristic Words of Each Topic

Topic	Characteristic Words
Self-Recording	record, diagnosis, treatment, effect, change, age, test, result, ...
Emotional Support	WeChat, difficult, hope, understanding, help, gratitude, pain, ...
Theoretical Knowledge	developmental disorder, basic research, genetic factors, intestinal microbiota, psychological mechanisms, ...
Medical Pathways	doctor, hospital, physician, prescription methods, institutions, recommendations, criticism, good intentions, cost, ...

**Figure 5: Emotion Value Scatter Distribution Diagram**

each sentiment value. The position and density of the points visually depict the emotional trend, from negative to positive. The plot approximately follows a normal distribution, reflecting the general emotional pattern.

The textual content of “Autism Bar” clearly demonstrates emotional characteristics. Negative emotions spread rapidly and have a broad impact, which requires attention and careful consideration. Negativity on the platform arises from several factors. First, the chronic stress faced by caregivers of children with autism—such as financial strain, social stigma, and physical exhaustion—often leads to anxiety, helplessness, and depression. Second, the lack of systematic support exacerbates these emotions, with many families relying on online communities as their primary outlet. Additionally, the promotion of highly discussed controversial content triggers an “emotional contagion” effect. This may ultimately shift discussions toward passivity and negativity, undermining the community’s supportive role.

6 Analysis and Summary

The social network structure of “Autism Bar” emphasizes the crucial role of opinion leaders and their significant influence on the community. Therefore, an online health community model centered on opinion leaders can be established, with a management system to regulate them, ensuring effective social guidance and macro control.

Text analysis categorizes the community based on users’ focus areas and interaction purposes, enhancing user retention. “Autism Bar” exhibits strong emotional characteristics, with emotional risks that may create tension and resistance. An interest group model can be developed, allowing users to self-organize, identify emotional trends, and facilitate emotional regulation..

7 Practical Interaction Design of the Online Health Community for Children with Autism

In summary, the interaction system of the online health platform for children with autism can be comprehensively designed from multiple perspectives, such as the opinion leader model, emotional support, and topic segmentation, to ensure that the platform can efficiently and attentively serve children with autism and their families. The design and functional architecture diagram of the online health community for children with autism is shown in Figure 6.

The specific functional and interaction design of the online health community for autism is as follows:

(1) Design based on the overall network structure characteristics of the autism online health community, the platform can adopt a group model design to leverage the clustering effect of social networks, bringing together users with similar interests. Social modules based on user preferences enhance the user experience and boost engagement.^[13] Users can create or join groups independently, with the platform also recommending relevant groups based on behavioral data. Group administrators help improve information quality and interaction efficiency.

(2) Design based on individual network characteristics, A multidimensional algorithm combining social position, content, and emotion identifies high-activity, high-influence users as opinion leaders. This study proposes a dual-track mechanism for opinion leader selection in online health communities, employing a human-machine collaborative approach. On one track, candidates are sourced through institutional recommendations and self-nomination, followed by manual review and professional assessment to verify practical experience and domain expertise. On the other track, the mechanism integrates Latent Dirichlet Allocation (LDA) topic modeling and sentiment analysis to enhance a modified PageRank algorithm for evaluating user influence. This addresses common issues such as topic ambiguity, emotional bias, and distorted influence metrics. The multidimensional algorithm dynamically adjusts influence scores based on users’ topic relevance and interaction depth, incorporating an emotion regulation factor (α) to respond in real time to shifts in community sentiment. By combining algorithmic ranking with human validation, the system effectively identifies the top 5% of high-quality opinion leaders, thereby facilitating knowledge dissemination and emotional support within autism-related health communities. The text is divided into K topics, with μ denoting the importance of subtopic k to the overall community, taking values in the range $[0, 1]$. Let $\theta_{u,k}$ represent the probability that node u belongs to topic k . Then, the global

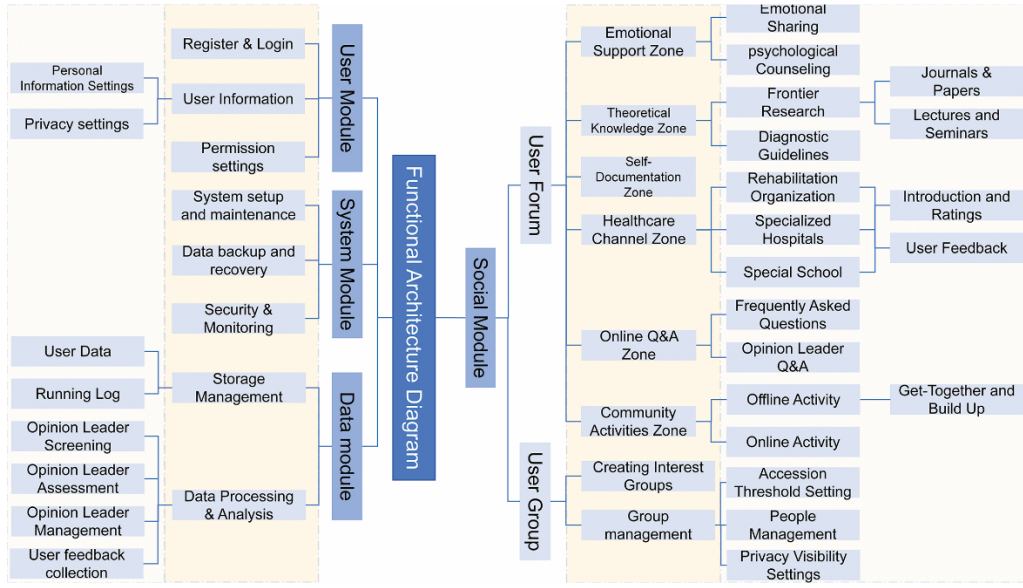


Figure 6: Functional Architecture Diagram of the Autism Online Health Platform

influence score of a connected node v is defined as:

$$S(v) = \mu \cdot PR_{\text{global}}(v) + (1 - \mu) \cdot \sum_{k=1}^K \theta_{u,k} \cdot PR_k(v) \quad (6)$$

To incorporate emotional regulation into the results, the sentiment value E is defined within the range of -1 to 1 . The final formula is expressed as follows:

$$S'(v) = S(v) \cdot (1 + \alpha \cdot E) \quad (7)$$

Regular feedback dynamically adjusts leader selection. Positions like moderators and “health ambassadors” are assigned to enhance engagement, with a reward system in place. Opinion leaders receive resources, including training and content tools, to boost influence and credibility. They are encouraged to share personal experiences, professional knowledge, and treatment advice, and participate in Q&A and online seminars to expand their positive impact.

(3) Design based on the text topic model:

The community is organized into topic modules, covering areas like theoretical knowledge, institution introductions, and emotional support. A clear labeling system allows users to easily find relevant information. An incentive system encourages participation, and high-quality posts are highlighted to increase visibility and engagement.

To mitigate negative sentiment, platforms can reshape community dynamics through multidimensional design. At the content level, balancing algorithms and emotional triage mechanisms—such as mandatory labeling, categorization, and adjusted recommendation weights—can limit negative content while promoting constructive posts like expert Q&As and rehabilitation cases. Incentives like visual badges may further encourage positive sharing. Using emotion management tools, such as AI-assisted diaries, can detect persistent negativity and suggest coping strategies like breathing exercises or counseling resources, guiding users from emotional release to problem-solving. A tiered support system can link relevant

resources to user posts and organize need-based groups (e.g., single-parent, school-age), aided by AI-generated discussion prompts and expert Q&A sessions to turn fragmented emotions into shared, structured knowledge. Setting an emotional feedback function, including likes, comments, and emojis, is implemented to encourage emotional expression. A dedicated emotional support section fosters mutual assistance and encourages positive emotional interaction, enhancing the community atmosphere.

(4) Design based on other directions:

Further refinement of the platform's interface design, user experience and community atmosphere can be enhanced through targeted features. For instance, an emotional labeling system guides content flow by requiring users to classify posts as help-seeking, neutral, or positive—visually coded in red, yellow, and green with reinforcing icons. The daily quick record function uses a smart “+” button to offer emoji check-in, voice shorthand, and template input. Parents can log key data (e.g., mood or training progress) in seconds, with entries timestamped into summary cards. These are processed into visual timelines showing mood trends, intervention overlap, and linked media. A pinch gesture activates comparison mode, displaying monthly interaction data side-by-side, while AI highlights developmental milestones. Selecting the red “Help” tab triggers a popup with local counseling resources. Topic-specific tabs and label filters, along with dynamic badges (e.g., an animated sunflower for every 100 clicks), incentivize constructive participation and reduce negative content overload through gamified interaction.

Given the community's sensitivity, privacy is ensured via a “double-mask” anonymous posting system, which hides user identity, location, and device data. Anonymous posts appear as abstract icons, revealing only current content while concealing history and social links. A “burn after reading” option allows uploaded images to auto-delete within 24 hours, with backend encryption preventing full data recovery even in case of leaks. These features offer

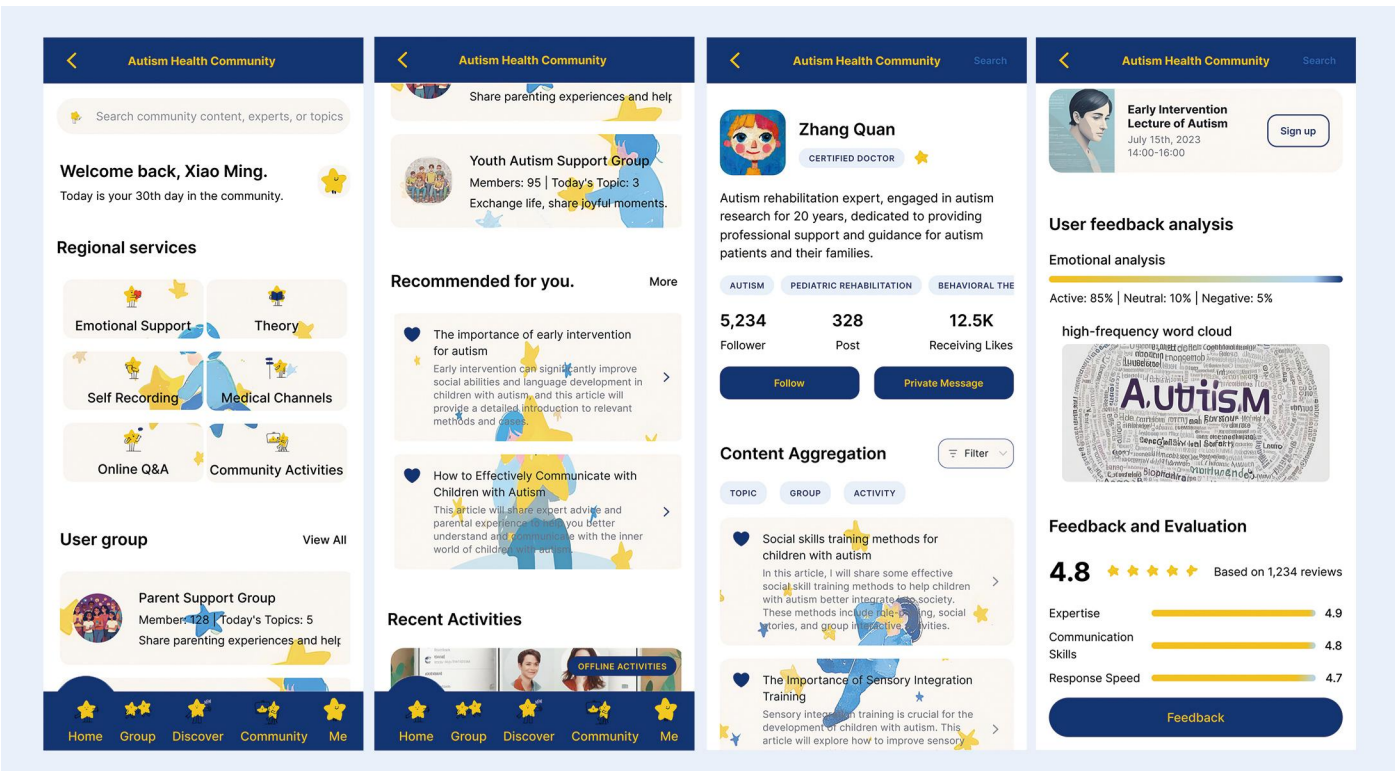


Figure 7: High-Fidelity Prototype Diagram of the APP Interface

emotional support while safeguarding privacy, enabling vulnerable users to engage safely and empathetically.

Color psychology is applied in the interface design, using warm and cool colors to highlight key areas. A minimalist approach ensures harmony in layout, text formatting, and background colors for a pleasant user experience. Calm blue and warm yellow are chosen for their therapeutic effects, promoting relaxation and alleviating negative emotions. The consistent color scheme enhances familiarity, comfort, and user satisfaction.

Additionally, the navigation menu should be intuitive, enabling users to quickly find needed functions and pages. Clear visual or auditory feedback for actions improves user control, reduces errors, and enhances efficiency^[14]. To enhance user experience, the platform should simplify task completion by reducing unnecessary steps. Personalized settings, such as adjustable layout and font size, will allow users to tailor the interface to their needs, improving convenience and satisfaction while accommodating diverse user preferences for ease of use.

Based on the design mentioned above, the design practice is carried out, and the high-fidelity prototype of the app interface is shown in Figure 7.

8 Conclusion

This paper conducts an in-depth analysis of the overall interaction system of Baidu Tieba's "Autism Bar" from two perspectives: social network and user behavior. The social network of Baidu "Autism Bar" shows a dense center with sparse edges, exhibiting a

small-world effect. There are numerous subgroups, but their connections are not tight, the topic classification is relatively clear, and the sentiment is predominantly negative. Based on the obtained characteristics, a functional interaction system design for the online health platform for children with autism has been developed. However, the overall platform design still has some shortcomings. The study only collected user interaction data from a single platform, and data from other small platforms can be introduced in the future to verify the theoretical accuracy. The reliability of the user influence algorithm remains to be validated. In the future, computational screening of large datasets can be conducted to compare the accuracy improvement of this algorithm with that of other methods. Further optimization of the platform could enhance program development by incorporating dynamic effects through computer programming and conducting effectiveness evaluations.

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